ELECTRONICS ENGINNERING

ELECTRONICS-II LABORATORY

2023-2024 FALL

EXPERIMENT-2

Single-BJT Amplifiers at Low and High Frequencies

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1 INTRODUCTION

Frequency response is the subject that shows how an electronic circuit or system responds to signals in frequency. The aim of this topic is to understand and evaluate how an electronic circuit processes signals in the frequency range and to use it in the most efficient frequency range. In this experiment, it was tried to observe how factors such as gains change by changing the frequencies of the established circuits.

1.1 The Basic Common-Emitter (CE) Circuit

1.1.1 The DC Situation

First, the base common emitter circuit is established. The DC analyze is made to find small signal parameters like β, α, re, rπ and gm. These parameters will be used for gain calculation.

1.1.2 Mid-Band Response

A sine wave signal of 1Vpp and 1Khz will applied to the node I and peak-to-peak voltages at nodes S, B, E, C, D. Here the gain of the signal can be observed.

1.2 Low frequency Response

1.2.1 Basic Overall Response

In this part of the experiment, starting by 1Khz the frequency is decreased, the gain is calculated by dividing vd and vs, and when the gain decrease by 3db this frequency will be noted.

1.3 High Frequency Operation of the Common-Emitter (CE) Amplifier

1.3.1 Basic AC Measurements:

This circuit is constructed to understand the behavior of the CE amplifier at high frequencies. Gains will be calculated by varying the frequencies. Based on theoretical knowledge, it is expected that the gain will decrease as the frequency increases. In the experiment, measurements will first be taken for f=10 kHz and then the frequency will be increased to test the theoretical information and a gain drop of 3dB will be observed. Measurements will be repeated with 10 times the frequency with 3 dB drop. The collector resistance will decrease with the resistor connected to the collector at d and the effect of this resistance will be observed by analyzing the circuit again.

2 Brief Theory and Theoretical Calculations

Provide a brief description of the subject and give your preliminary results (calculations, simulations etc.) (Use Abadi with 12fontsize)

2.1 The Basic Common-Emitter (CE) Circuit

2.1.1 Theorical information

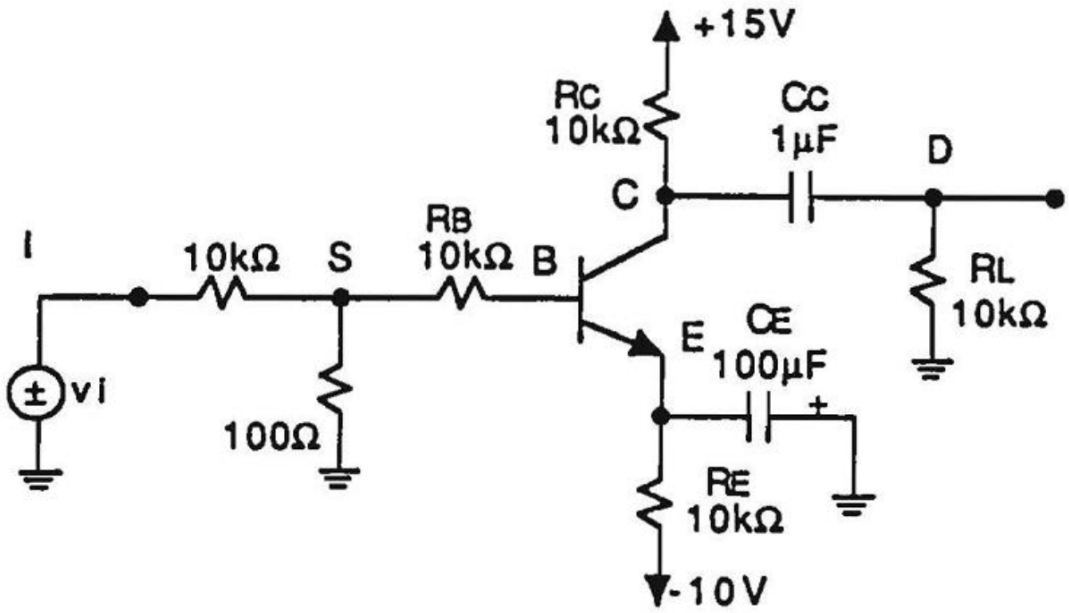


Figure 1: A basic CE BJT

In BJT, the small signal parameters can be defined by the follow:

$$\beta = \frac{I_C}{I_B}$$

$$\alpha = \frac{\beta}{\beta + 1}$$

$$gm = \frac{I_C}{V_T}$$

$$r_\pi = \frac{\beta}{gm}$$

$$r_e = \frac{\alpha}{gm}$$

Using this parameter, the small signal gain can be calculated. If the circuit is simplified for small signal analyze, by analyzing in the mid band of frequency, the capacitors can be accepted as short circuit so the circuit will be as Figure 2.

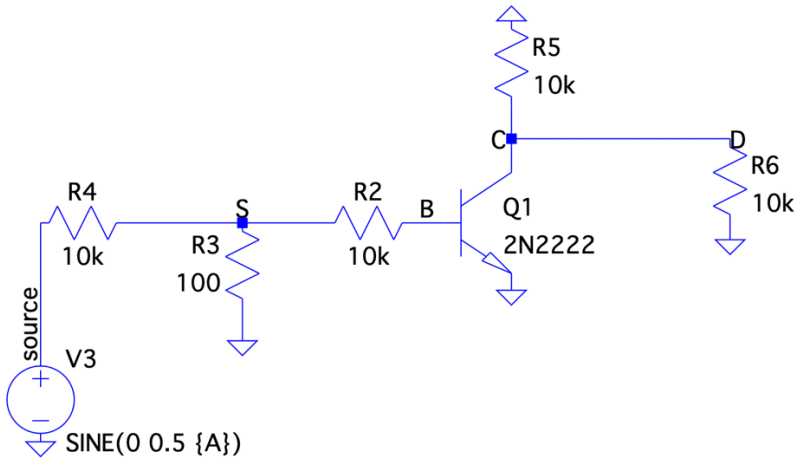


Figure 2: Simplified CE circuit for ac analyze (mid band)

From the simplified circuit Vc/Vb = - gm (R5 // R6) = -gm (R_C // R_L) = -gm (5k) it is the gain of the signal applied from the base.

2.1.2 The DC Situation

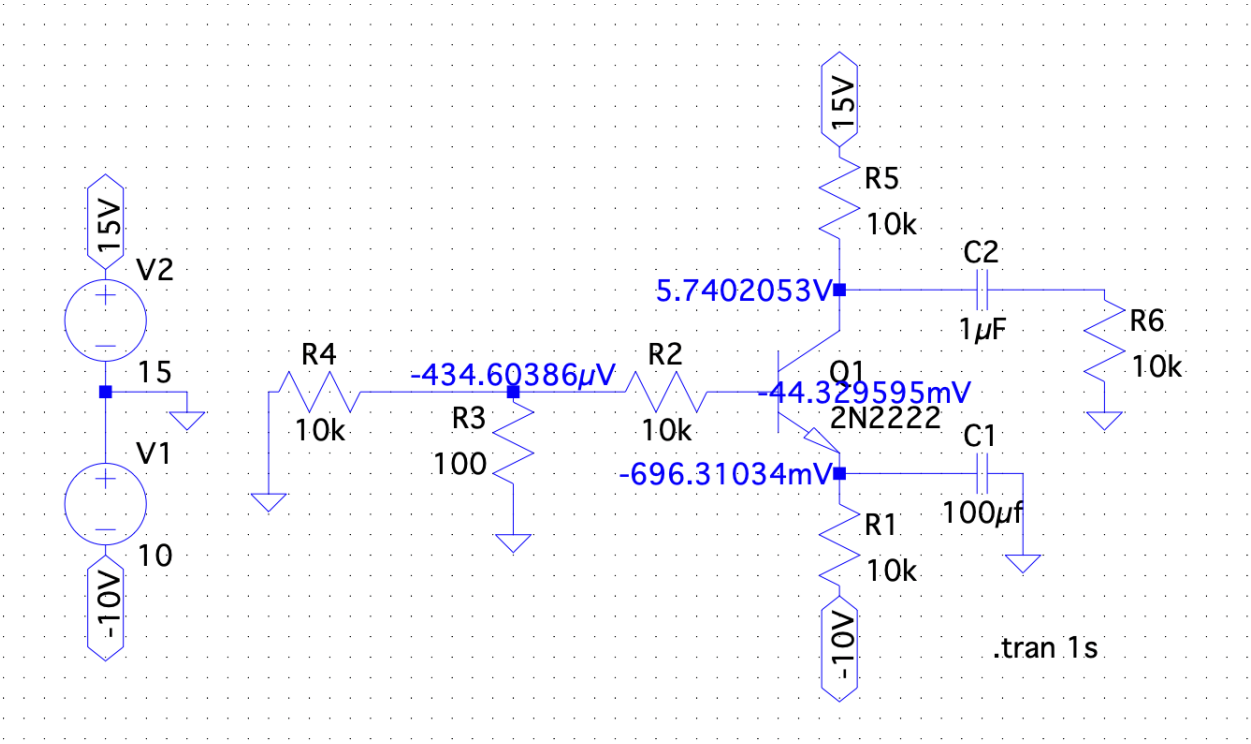


Figure 3: LTspice simulation for DC values for CE BJT amplifier circuit

V_S	V_B	V_E	V_C	I_E	I_C
-0.434 mV	-44.33mV	-696.31mV	5.74V	0.930 mA	0.926 mA
I_B	β	α	r_e	r_π	g_m
0.0044 mA	210.4	0.995	26.87ohm	5.68 kohm	37.04 mS

Table 1: DC values for CE BJT amplifier circuit from simulation

$$I_E = \frac{V_E - (-10\ V)}{10kohm} = 0.930\ mA$$

$$I_C = \frac{15V - V_C}{10kohm} = 0.926\ mA$$

$$I_B = \frac{Vs - V_B}{10kohm} = 0.0044\ mA$$

$$\beta = \frac{I_C}{I_B} = 210.4$$

$$\alpha = \frac{\beta}{\beta + 1} = 0.995$$

$$gm = \frac{I_C}{V_T} = \frac{0.926mA}{25mV} = 0.03704\ S = 37.04\ mS$$

$$r_\pi = \frac{\beta}{gm} = 5680\ ohm = 5.68\ kohm$$

$$r_e = \frac{\alpha}{gm} = \frac{0.995}{37.04 * 10^{-3}} = 26.87\ ohm$$

2.1.3 Mid-Band response

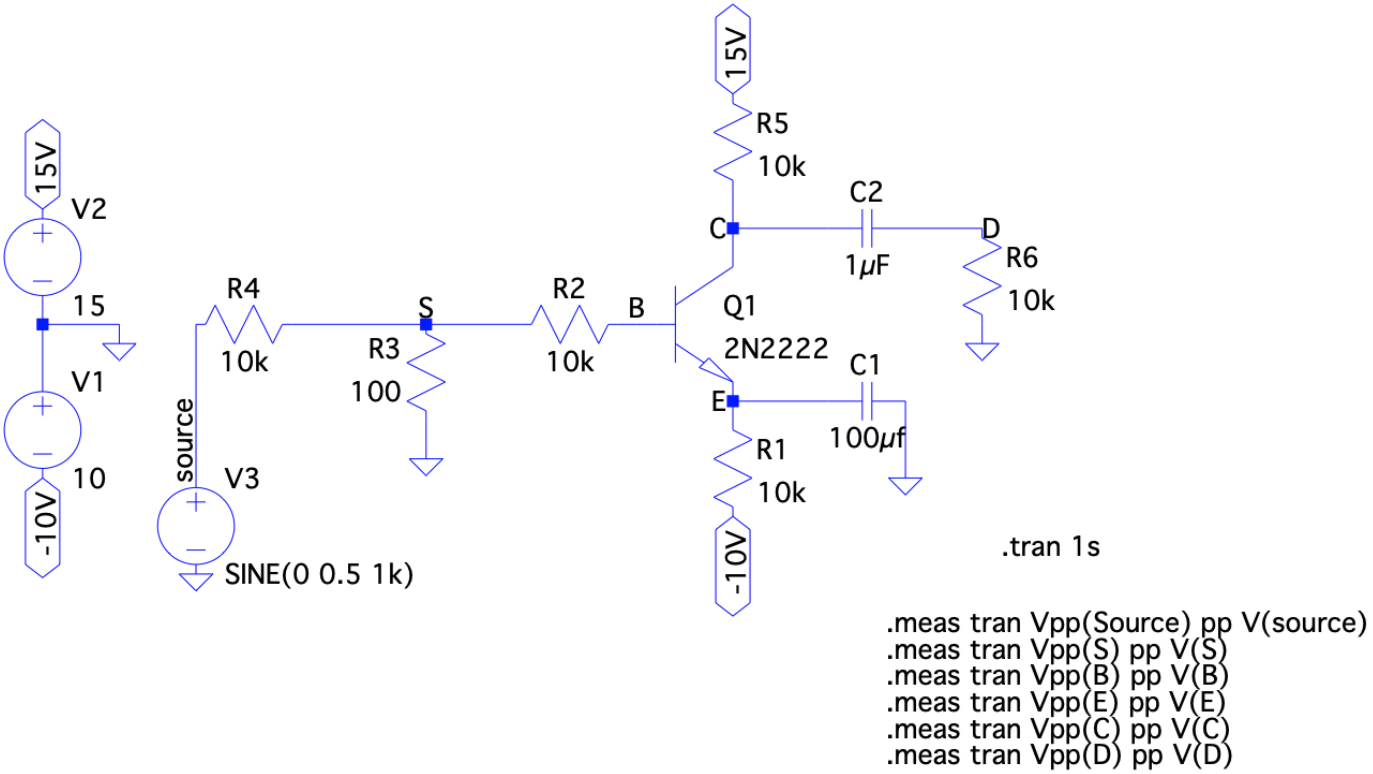


Figure 4:CE BJT amplifier circuit with 1 Vpp at 1khz

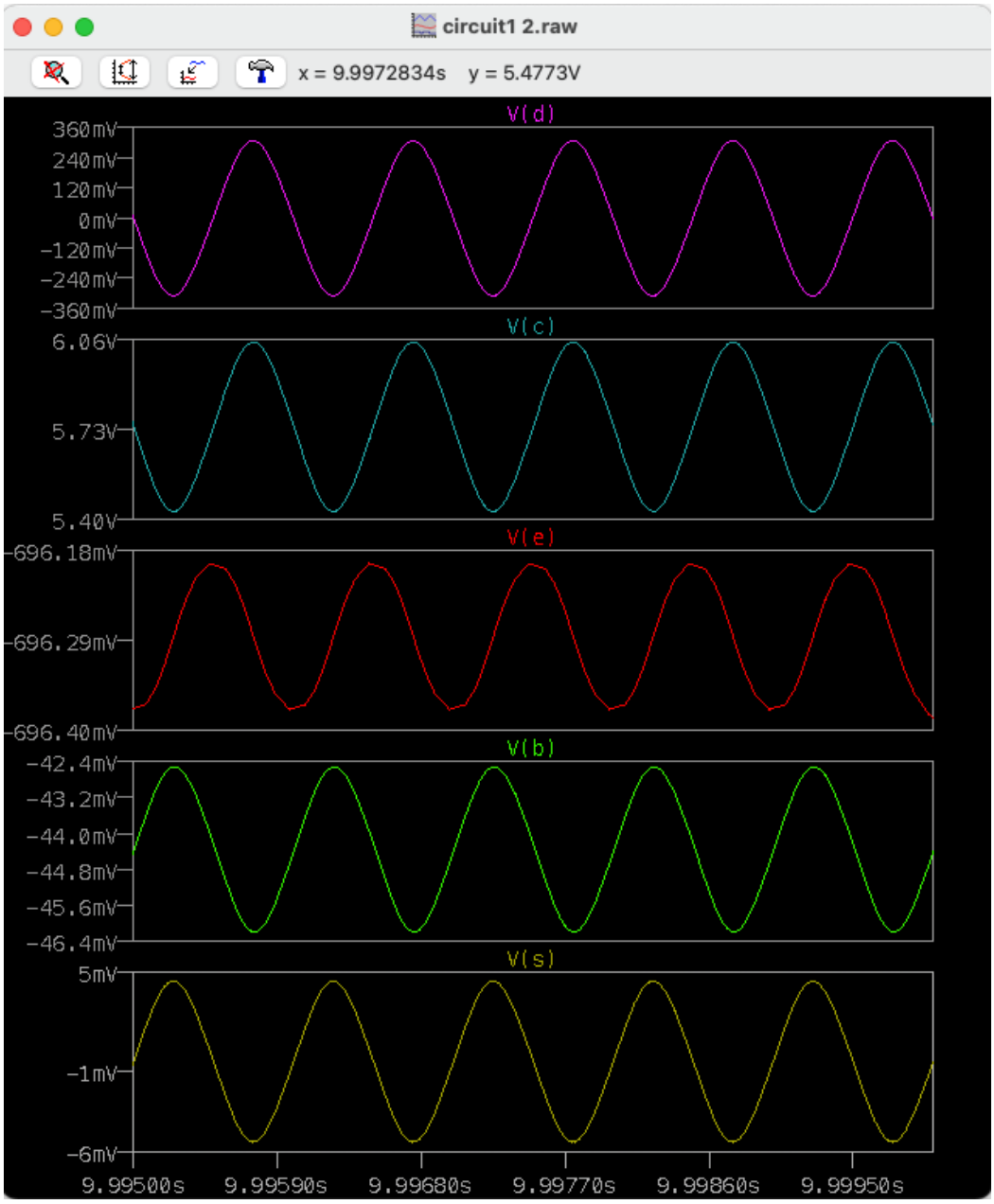


Figure 5: The graph of each voltage of circuit in Figure 2

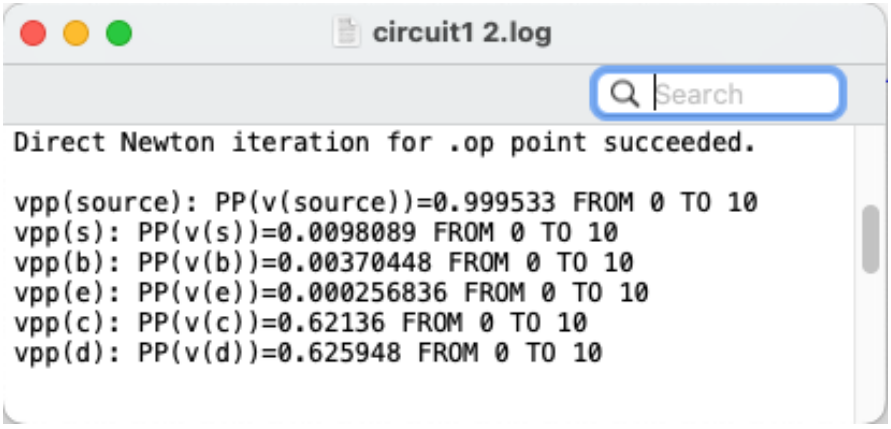


Figure 6: Log of LTspice that includes peak to peak voltages

us	ub	ue	uc	ud
9.81mV	3.70mV	0.26mV	0.62V	0.626 V

Table 2: Peak-to-peak values for CE BJT amplifier circuit from simulation

2.2 Low Frequency Response

2.2.1 Basic Overall Response

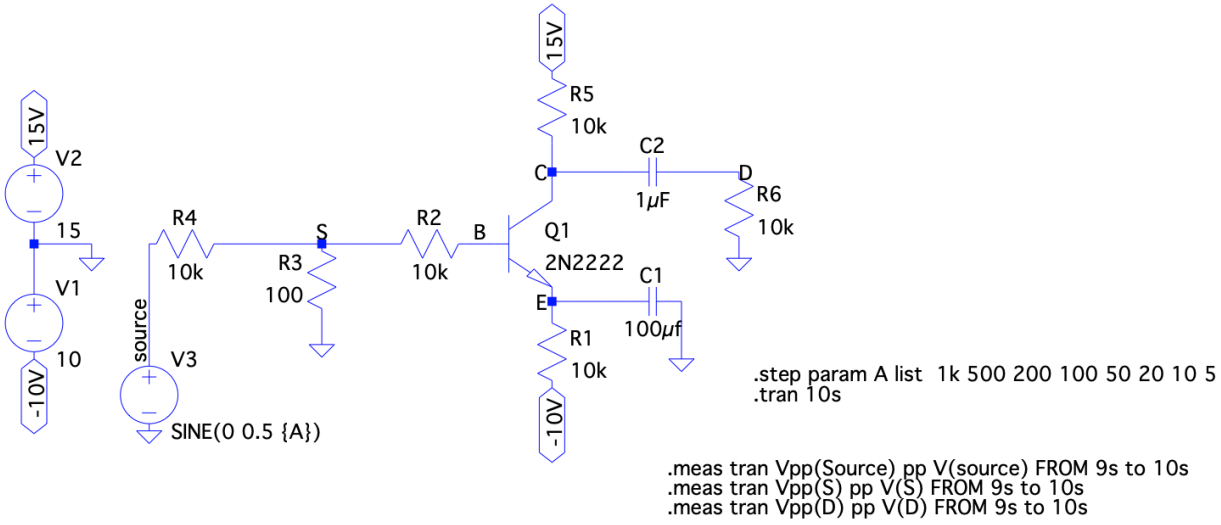


Figure 7: CE BJT Amplifier circuit and simulation for multiple frequencies

circuit1 2.log

```
.step a=5
.step a=10
.step a=20
.step a=50
.step a=100
.step a=200
.step a=500
.step a=1000


Measurement: vpp(source)
  step           PP(v(source))    FROM            TO
     1           0.999998         9               10
     2           0.999993         9               10
     3           0.999533         9               10
     4           0.999533         9               10
     5           0.999533         9               10
     6           0.999533         9               10
     7           0.999533         9               10
     8           0.999533         9               10

Measurement: vpp(s)
  step           PP(v(s))         FROM            TO
     1           0.00987284       9               10
     2           0.00986273       9               10
     3           0.00983773       9               10
     4           0.00981635       9               10
     5           0.00981056       9               10
     6           0.00980897       9               10
     7           0.00980856       9               10
     8           0.0098085        9               10

Measurement: vpp(d)
  step           PP(v(d))         FROM            TO
     1           0.0777159        9               10
     2           0.206946         9               10
     3           0.386001         9               10
     4           0.564177         9               10
     5           0.604749         9               10
     6           0.616342         9               10
     7           0.619142         9               10
     8           0.619519         9               10
```

Figure 8: LTspice log for peak to peak voltages for vairent frequencies

	us	ud	ud/ us
1 kHz	0.0098085	0.619519	63.16144
500 Hz	0.00980856	0.619142	63.12262
200 Hz	0.00980897	0.616342	62.83453
100 Hz	0.00981056	0.604749	61.64266
50 Hz	0.00981635	0.564177	57.4732
20 Hz	0.00983773	0.386001	39.2368
10 Hz	0.00986273	0.206946	20.98263
5 Hz	0.00987284	0.0777159	7.87169

Table 3: Peak to peak voltages for corresponding frequencies

F_L	20 Hz

Table 4: Cutoff frequency

And using Ac sweap simulation and by dividing Vd to Vs the gain is obtained as a function of the frequency. It is clear from the graph that the gain drops by 3 db when the frequency is around 20 hz.

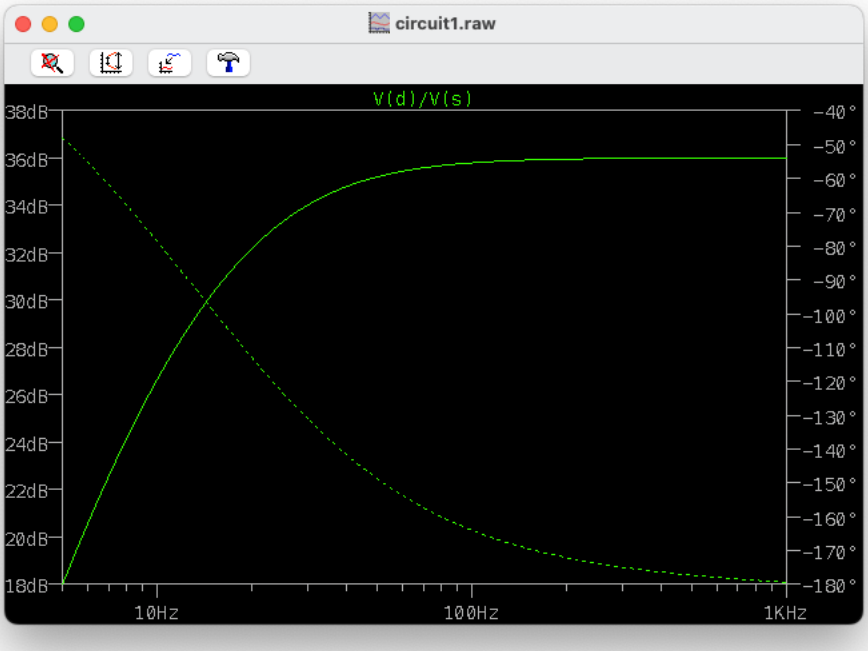


Figure 9: AC sweep simulation result, The bode plot of the gain

2.3 High Frequency Operation of the Common-Emitter (CE) Amplifier

2.3.1 Theorical information

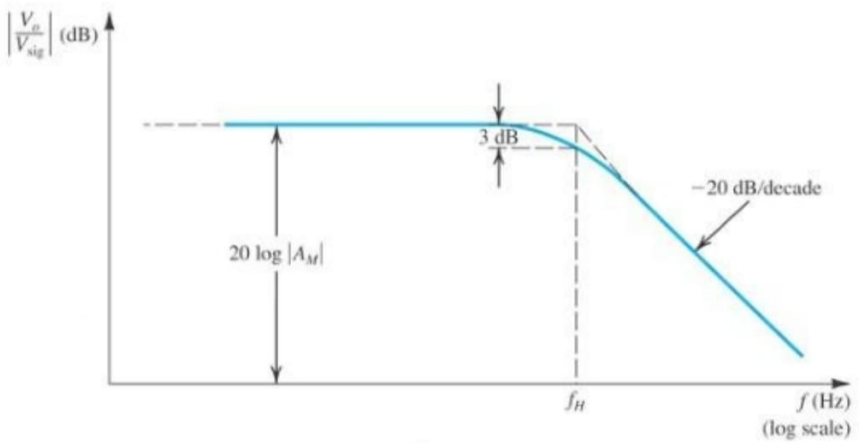


Figure 10: CE Amplifier High frequency responce

In CE circuits, a gain drop is expected at high frequency as shown in the figure. Considering the formula Vc/Vb=-gm×Rc in CE circuits, the gain is expected to decrease as Rc decreases.

2.3.2 Basic AC Measurements:

Rc	Input	f	Vs	Vb	Vc
10 kΩ	10 kHz	10 kHz	0.039 V	0.012 V	2.834 V
	f1	45 kHz	0.039 V	0.008 V	2.014 V
	10f1	450 kHz	0.039 V	0.001 V	0.279 V
5 kΩ	10 kHz	10 kHz	0.039 V	0.016 V	2.356 V
	f1	65 kHz	0.039 V	0.001 V	1.64 V
	10 f1	650 kHz	0.039 V	0.001 V	0.225 V

Table 5:Peak-to-peak values and gains for two cases.

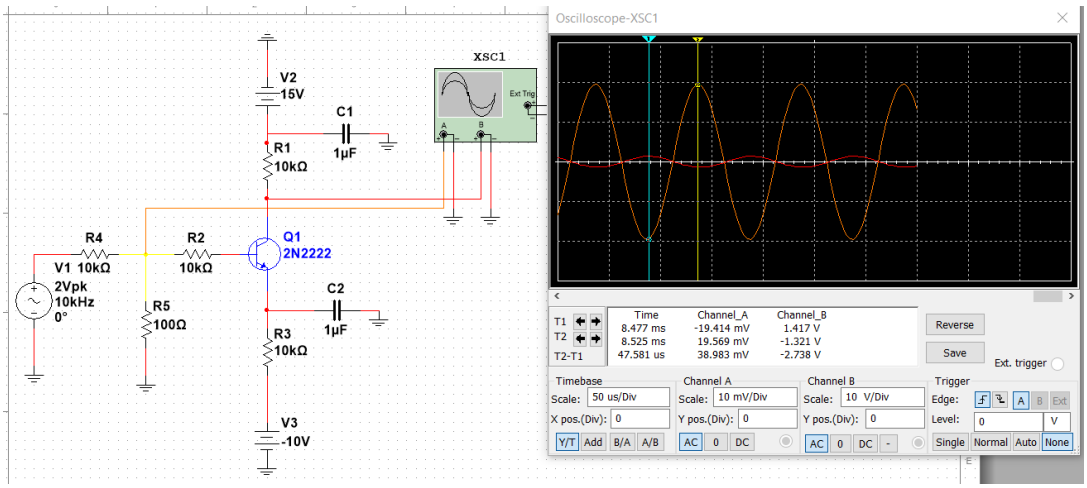


Figure 11:Vs and Vc simulation results for Rc=10kΩ,f=10kHz

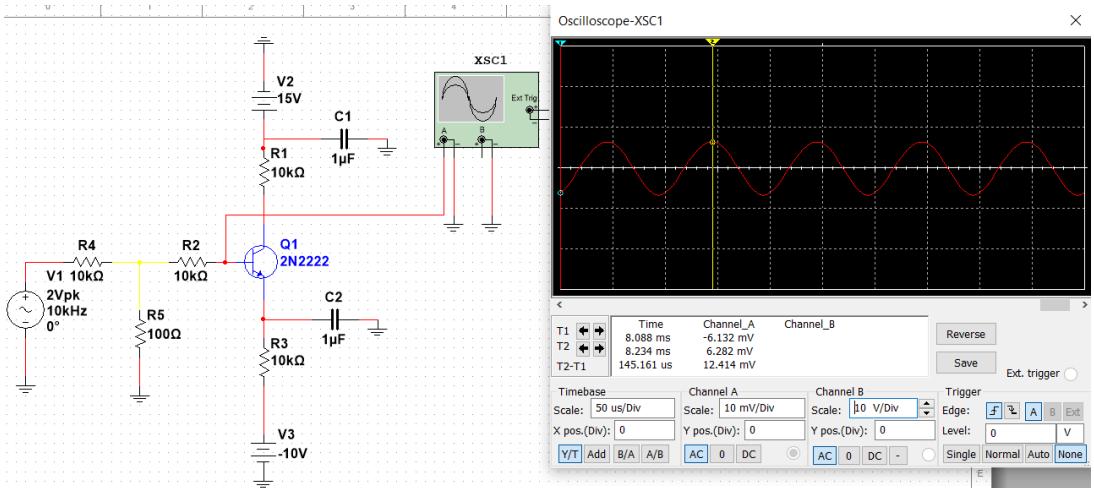


Figure 12: Vb simulation results for Rc=10kΩ,f=10kHz

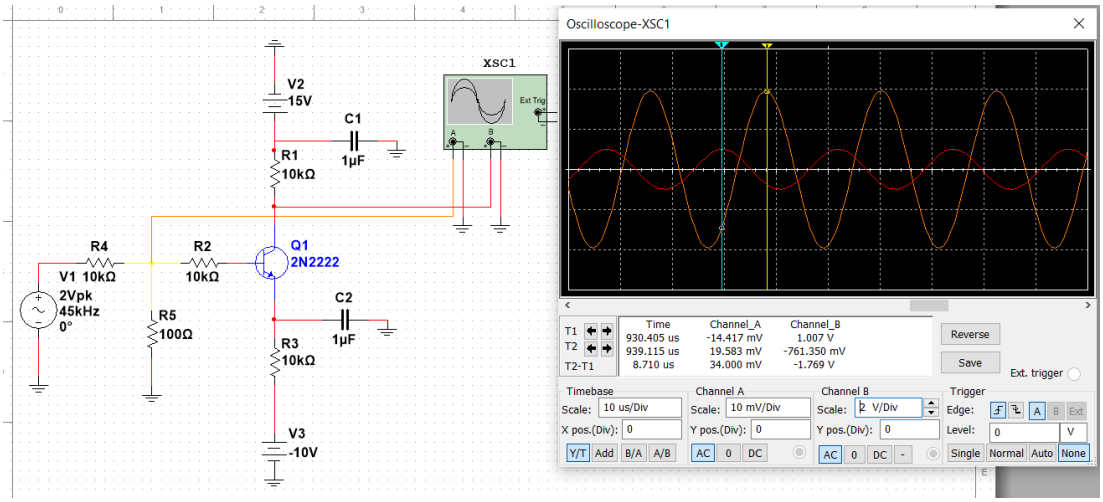


Figure 13: Vs and Vc simulation results for Rc=10kΩ,f=45kHz

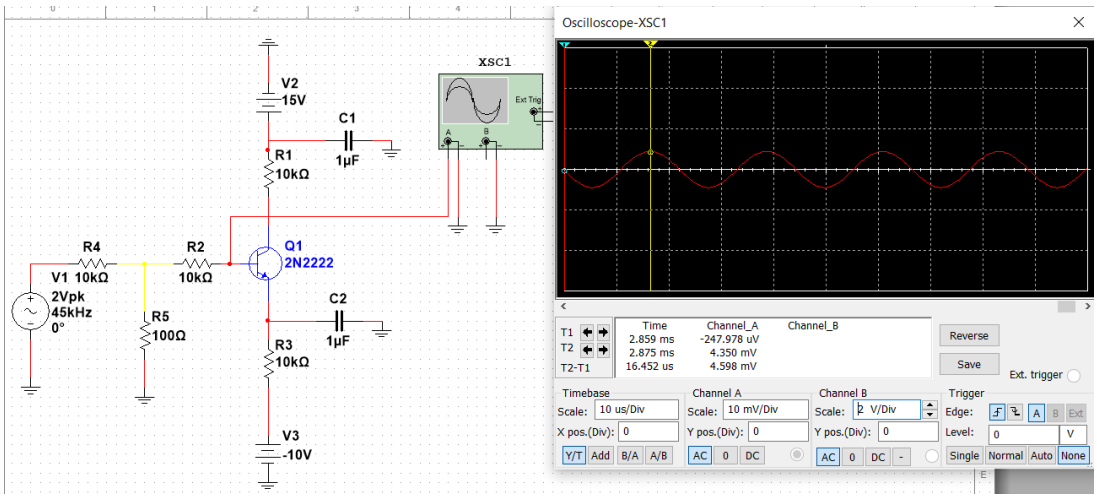


Figure 14: Vb simulation results for Rc=10kΩ,f=45kHz

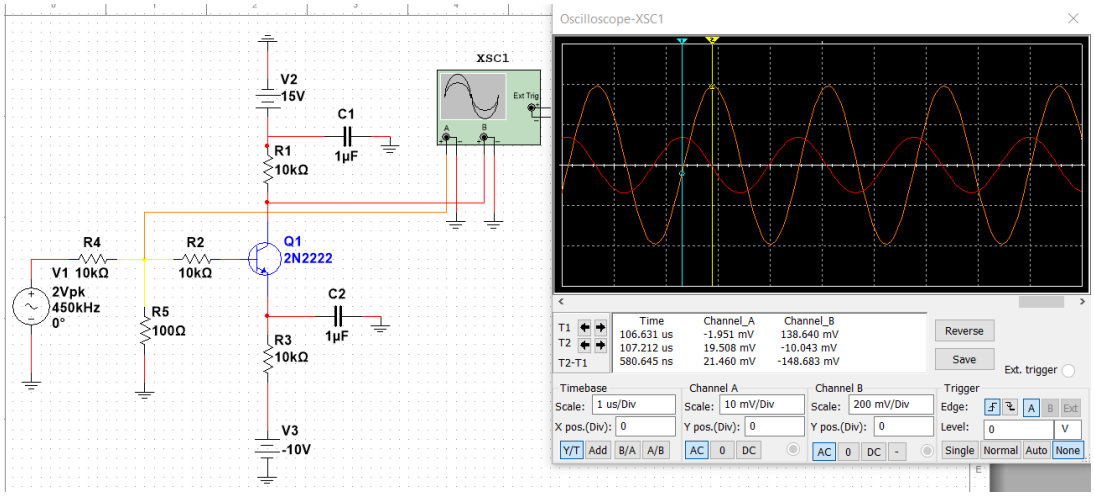


Figure 15: Vs and Vc simulation results for Rc=10kΩ,f=450kHz

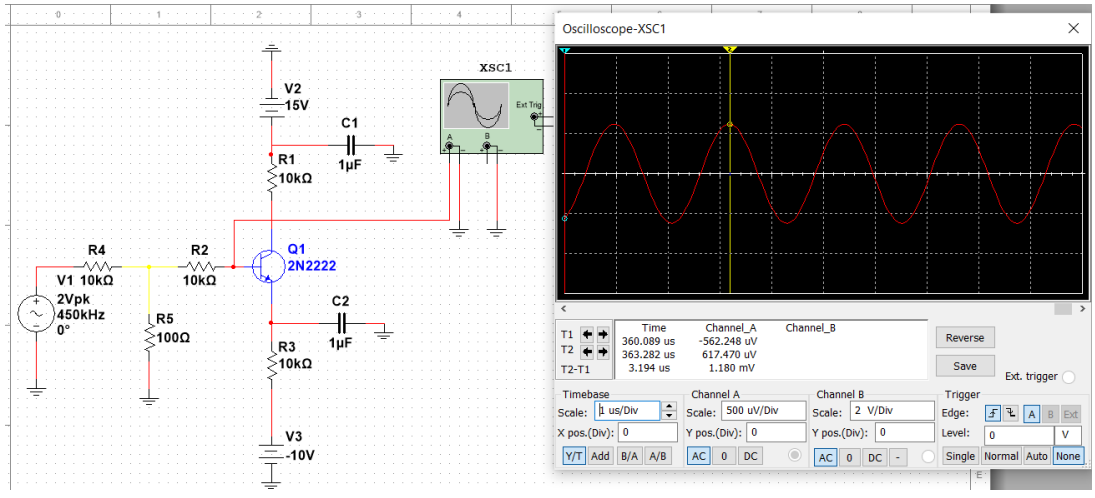


Figure 16: Vb simulation results for Rc=10kΩ,f=450kHz

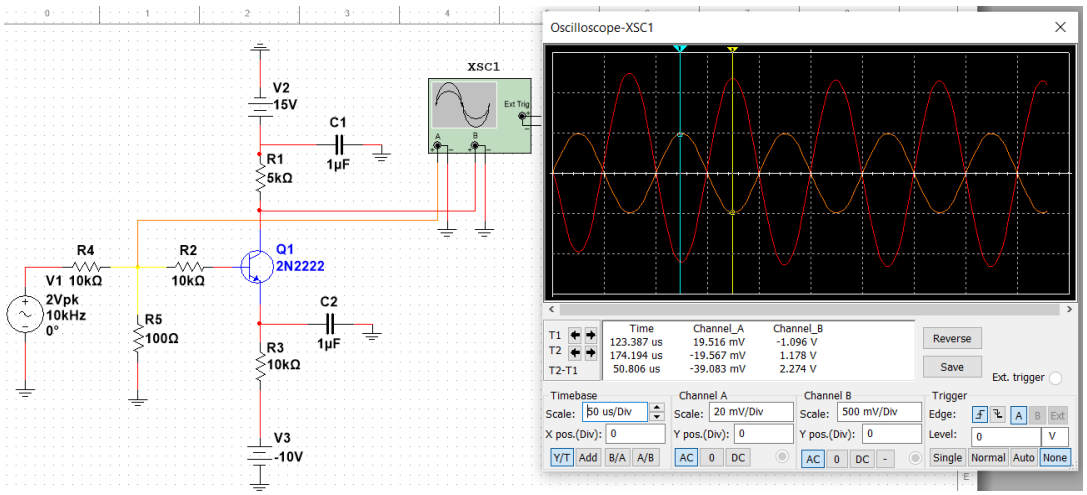


Figure 17: Vs and Vc simulation results for Rc=5kΩ,f=10kHz

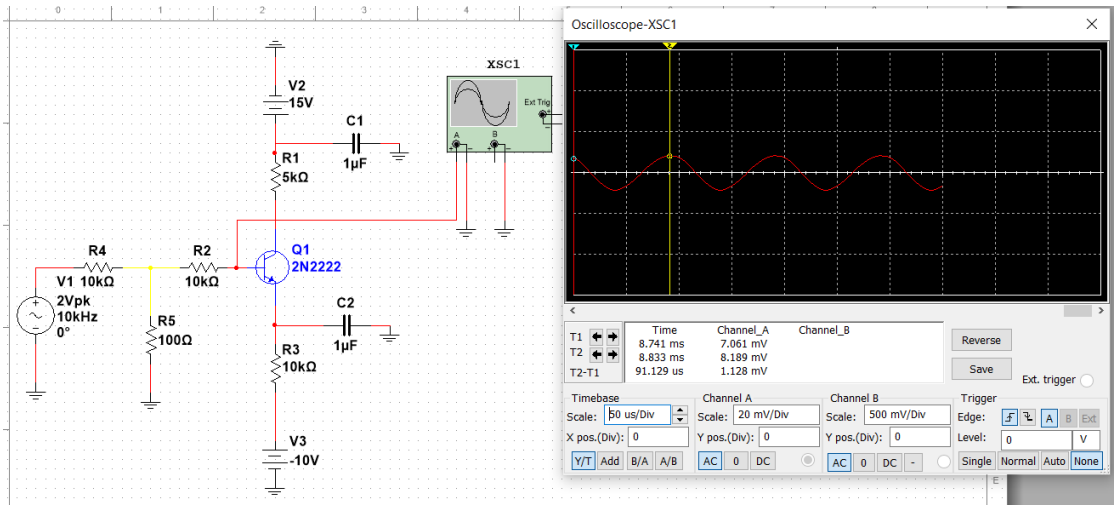


Figure 18: Vb simulation results for Rc=5kΩ,f=10kHz

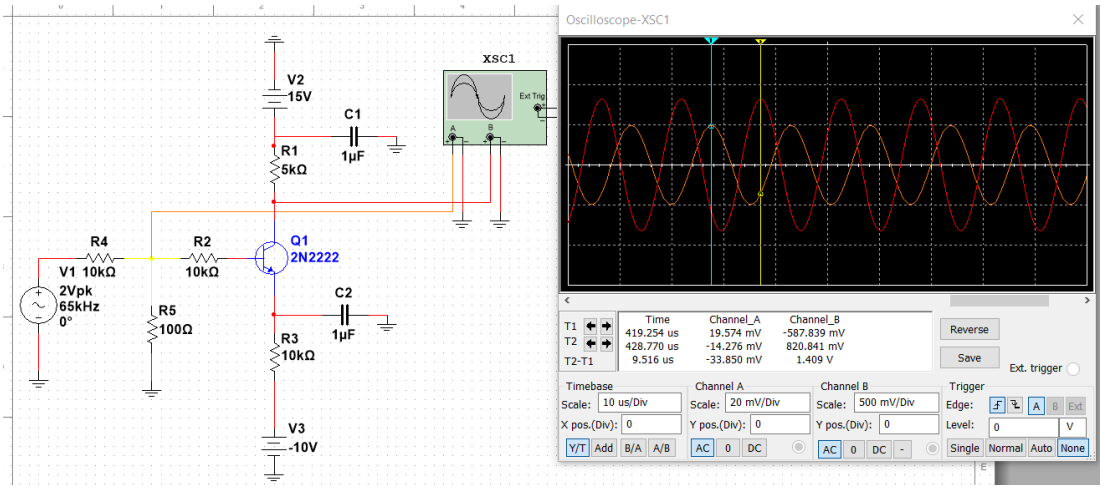


Figure 19: Vs and Vc simülation results for Rc=5kΩ,f=65kHz

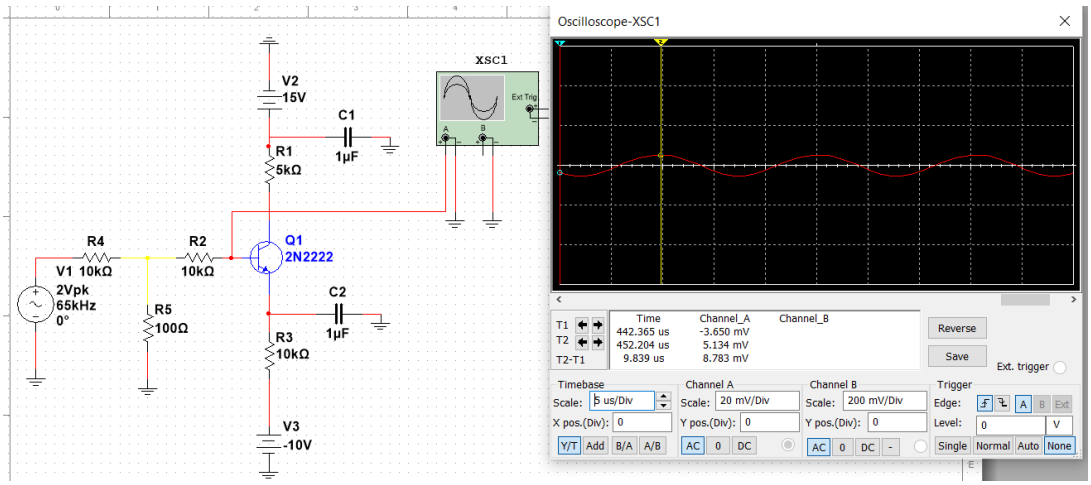


Figure 20:Vb simülation results for Rc=5kΩ,f=65kHz

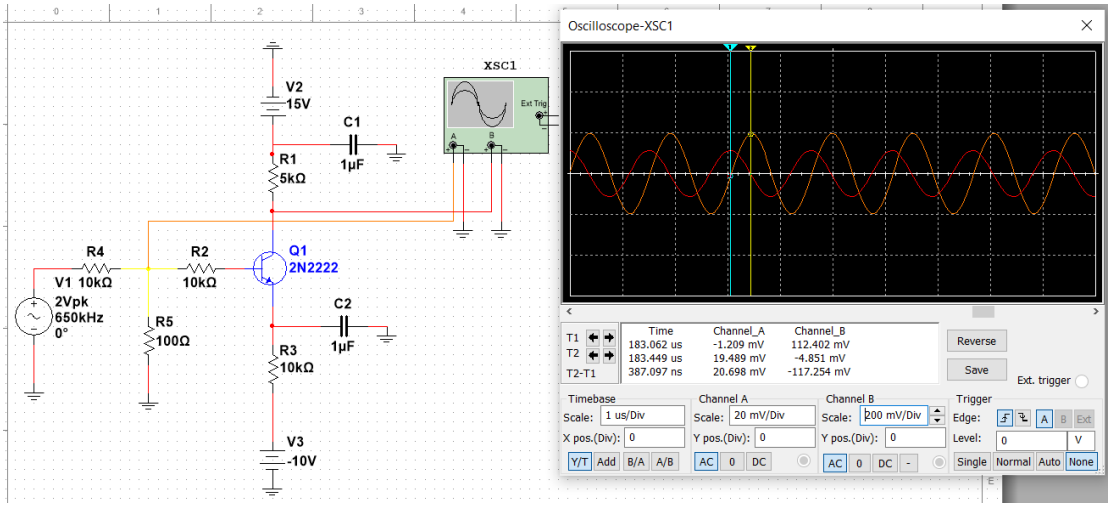


Figure 21: Vs and Vc simülation results for Rc=5kΩ,f=650kHz

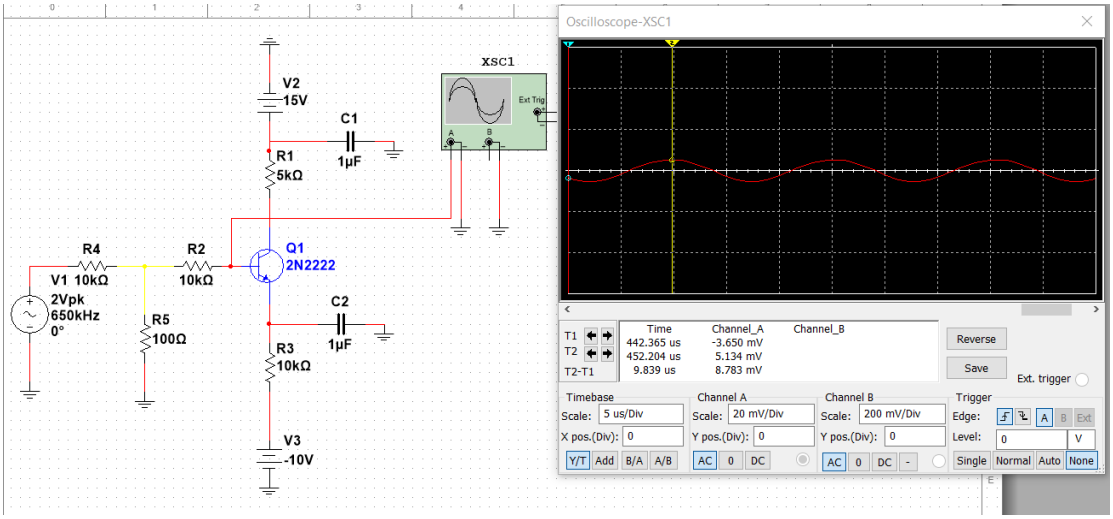


Figure 22: Vb simülation results for Rc=5kΩ,f=650kHz

For Rc=10kΩ:

For 10 Khz: $Gain = 20 \times log(Vc \div Vs) = 20 \times log(2.834 \div 0.039) = 37.22\ dB$

If raise the frequency, until the voltage at C falls by 3 dB, for Vc=2.014 V, the frequency is 45 Khz and the gain is:

$Gain = 20 \times log(Vc \div Vs) = 20 \times log(2.014 \div 0.039) = 34.25\ dB$

For Rc=5kΩ:

For 10 Khz:

$Gain = 20 \times log(Vc \div Vs) = 20 \times log(2.356 \div 0.039) = 35.62\ dB$

If raise the frequency, until the voltage at C falls by 3 dB,for Vc=1.64 V, the frequency is 65 Khz:

$$Gain = 20 \times log(Vc \div Vs) = 20 \times log(1.64 \div 0.039) = 32.48\ dB$$

3 Experimental Results (Figures, tables, graphs)

3.1 The basic common emitter circuit

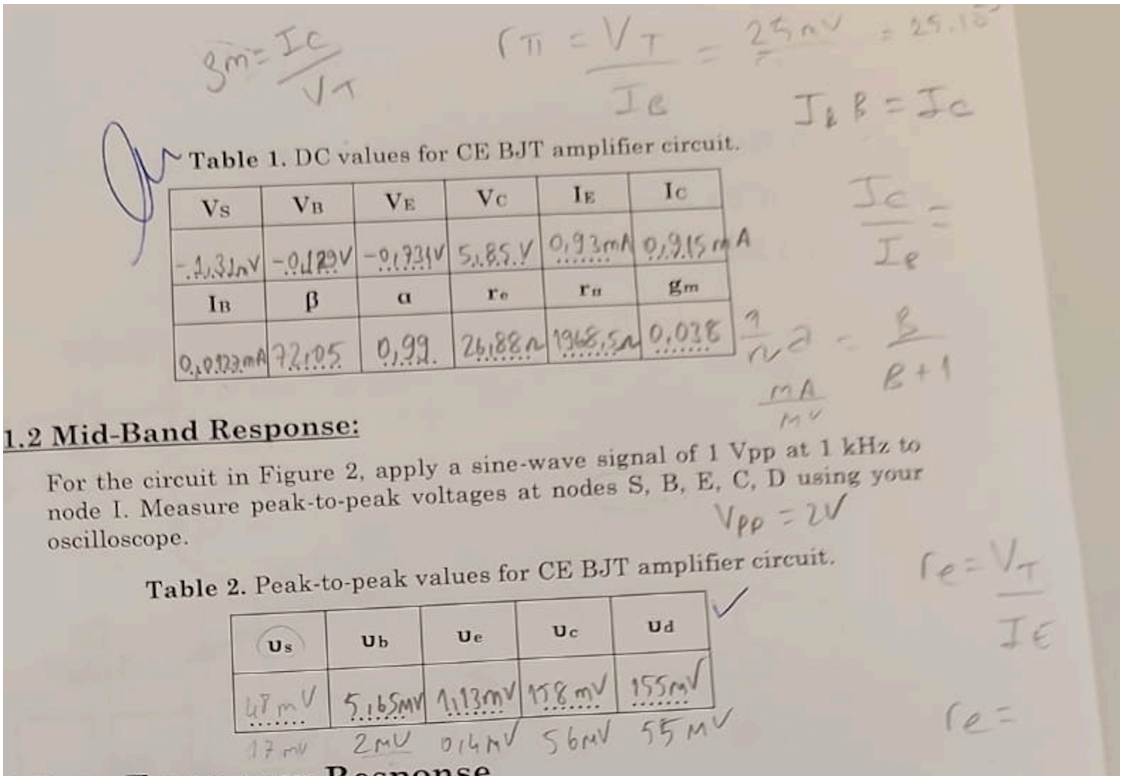


Table 1. DC values for CE BJT amplifier circuit.

V_S	V_B	V_E	V_C	I_E	I_C
-1.31mV	-0.129V	-0.731V	5.85V	0.93mA	0.915mA
I_B	β	α	r_e	r_π	g_m
0.0123mA	72.105	0.99	26.88Ω	1968.5Ω	0.038

1.2 Mid-Band Response:

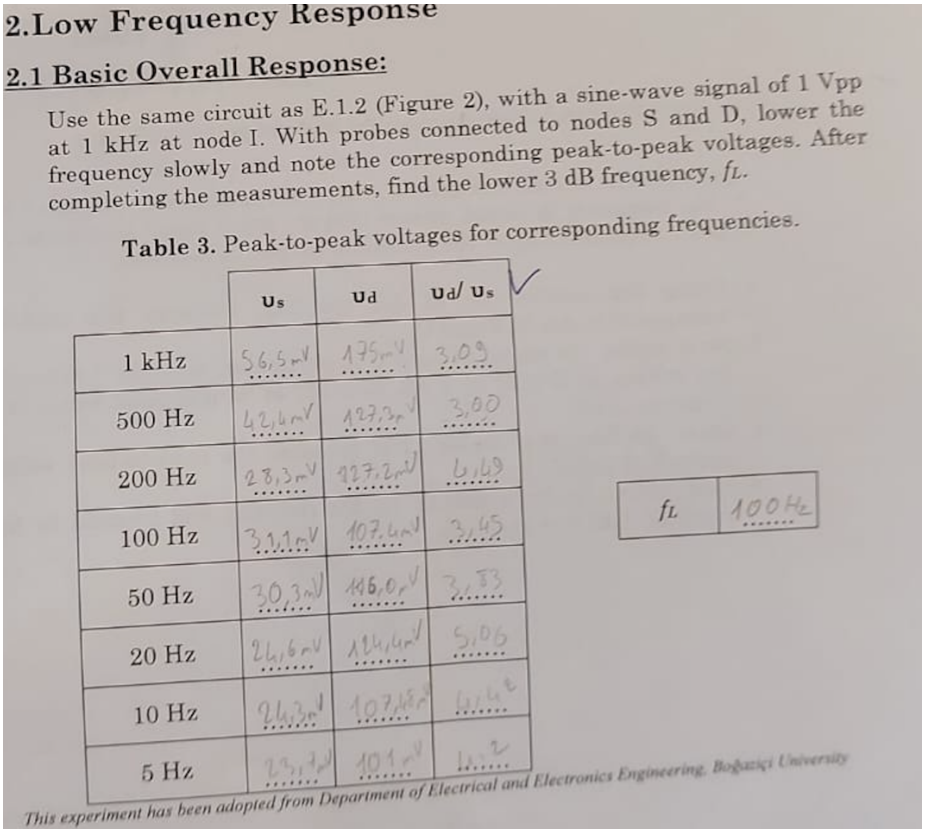
For the circuit in Figure 2, apply a sine-wave signal of 1 Vpp at 1 kHz to node I. Measure peak-to-peak voltages at nodes S, B, E, C, D using your oscilloscope.

Table 2. Peak-to-peak values for CE BJT amplifier circuit.

U_s	U_b	U_e	U_c	U_d
47mV	5.165mV	1.13mV	158mV	155mV

Figure 23: The common emitter circuit dc values and mid band response

From the dc analyze β value is seen as 72, it is because the transistor not connected success and it's reverse biased.



2.Low Frequency Response

2.1 Basic Overall Response:

Use the same circuit as E.1.2 (Figure 2), with a sine-wave signal of 1 Vpp at 1 kHz at node I. With probes connected to nodes S and D, lower the frequency slowly and note the corresponding peak-to-peak voltages. After completing the measurements, find the lower 3 dB frequency, f_L.

Table 3. Peak-to-peak voltages for corresponding frequencies.

	v_s	v_d	v_d / v_s
1 kHz	56,5mV	175mV	3,09
500 Hz	42,4mV	127,2mV	3,00
200 Hz	28,3mV	177,2mV	6,49
100 Hz	31,1mV	107,4mV	3,45
50 Hz	30,3mV	116,0mV	3,83
20 Hz	24,6mV	124,4mV	5,06
10 Hz	24,3mV	107,8mV	4,44
5 Hz	23,1mV	101,1mV	4,38

f_L	100Hz

This experiment has been adopted from Department of Electrical and Electronics Engineering, Boğaziçi University

Figure 24: Low frequency response peak to peak values

3.2 High Frequency Operation of the Common-Emitter (CE) Amplifier

3.2.1 Basic AC Measurements:

During the experiment, data could not be collected due to time constraints.

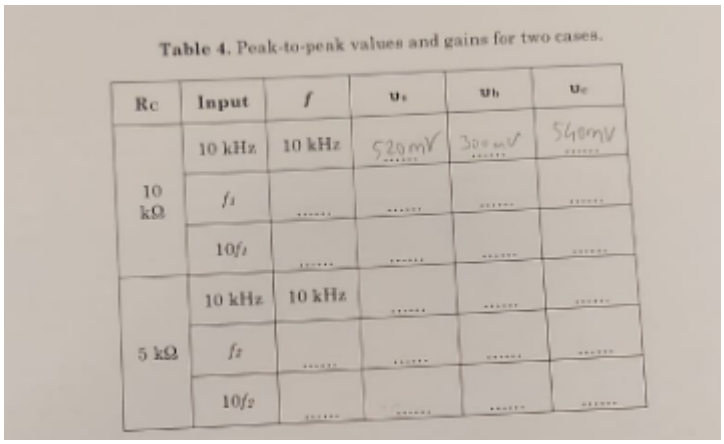


Table 4. Peak-to-peak values and gains for two cases.

R_C	Input	f	v_s	v_b	v_c
10 kΩ	10 kHz	10 kHz	520mV	300mV	540mV
	f_1				
	$10f_1$				
5 kΩ	10 kHz	10 kHz			
	f_2				
	$10f_2$				

Figure 25: High Frequency Operation of the Common-Emitter

4 Discussion

During the experiment, the npn transistor is connected in reverse because the provided connection in the experiment sheet is a bottom view but we looked to it as top view. So, the data is not collected true. Therefore, in this section, comments are made according to the simulation.

4.1 The Basic Common-Emitter (CE) Circuit

4.1.1 Mid-Band Response

The gain from the base is defined as Vc/Vb = - gm (R5 // R6) = -gm (R_C // R_L) = -gm (5k). And gm= $37.04\ mS$ so Vc = Vb*(37.04)*(5)= (3.7*10^-3)*(37.04)*(5)= 0.685V and from the simulation Vc obtained as 0.62V from the simulation so it's near to the voltage finded from the formula.

4.2 Low Frequency Response

The decrease in gain at frequencies under a positive point is called the roll-off. The frequency at which the gain begins to decrease notably is frequently called the lower cutoff frequency or -3dB frequency. Below this frequency, the benefit decreases at a rate of approximately 20 dB according to decade. From the simulation the cut-off frequency is seen as 20 Hz. The coupling capacitor capacitors are used to couple the amplifier with the signal source and the load. At low frequencies, the reactance of these capacitors increases, which reduces the overall gain.

4.3 High Frequency Operation of the Common-Emitter (CE) Amplifier

4.3.1 Basic AC Measurements:

Gain was measured by increasing the frequency values. In particular, the frequency at which the gain value at 10kHz decreased by 3dB was tried to be found. It was observed that the gain decreased as the frequency increased. The measurements were repeated when Rc=5kohm. Based on the generalized CE circuit, it is seen that gain decreases when Rc decreases as seen in the formula Vc/Vb=-gm×Rc.

5 References

[1]Adel S. Sedra and Kenneth C. Smith,Microelectronic Circuits,2015,SEVENTH EDITION